UNIVERSITY OF THE PHILIPPINES

STORM SURGE MODELING IN THE PHILIPPINE SETTING USING ADCIRC+SWAN THROUGH PARALLEL COMPUTING

by

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An Undergraduate Research Project Submitted to the

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Department of Computer Science

This undergraduate research project hereto attached, entitled **Storm Surge Modeling in the Philippine Setting Using ADCIRC+SWAN Through Parallel Computing**, prepared and submitted by **Joshua Frankie Rayo**, **Ryan Anthony Jacinto** and **Tricia Mae Esguerra**, in partial fulfillment of the requirement for the degree of **Bachelor of Science in Computer Science**, is hereby accepted.

Vena Pearl Bongolan, Ph.D.
Adviser

This undergraduate research project is hereby officially accepted and approved as partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science**.

Prospero Naval, Ph.D.
Chair
Department of Computer Science

UNIVERSITY OF THE PHILIPPINES

Bachelor of Science in Computer Science
(Disaster Risk Modeling and Simulation)

Joshua Frankie Rayo, Ryan Anthony Jacinto and Tricia Mae Esguerra

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We hereby declare that we have created this work completely on our own and used no other sources or tools than the ones listed, and that we have marked any citations accordingly.

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Most of all, to the Lord Almighty who wants the best for us and leads us to places where we really belong.

Abstract

Storm Surge Modeling in the Philippine Setting Using ADCIRC+SWAN Through Parallel Computing

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Ryan Anthony Jacinto
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University of the Philippines, 2015

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The Philippines is uniquely very archipelagic, and, relatively, the Philippines experience a higher number of typhoons. Consequently, devastating storm surges occurred all over the country, though it is only through the recent Typhoon Yolanda that it is once more brought to common knowledge. No model nor software has been tailored specifically for the Philippine setting to measure storm surges that occur every time a typhoon is in the county. Thus, the researchers are tailoring a model specifically for the Philippines and designing it to compute fast enough to be able to give real-time storm surge predictions around the country.

The model being used for this is the ADCIRC (ADvanced Multi-Dimensional CIRCulation) model coupled with SWAN (Simulating WAves Nearshore) model. The SWAN model employs an analog of the Gauss-Seidel sweeping method, while the ADCIRC model employs a continuous-Galerkin solution of the Generalized Wave Continuity Equation.

In order to not sacrifice accuracy for speed which was designed for real-time use, the use of parallel computing was employed so as to be able to more efficiently compute using the model. This was done through a 10-node parallel computer network through a Beowulf cluster built using 10 units of used AMD64 boxes (Intel x86_64 architecture).

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Chapter 1

Introduction

1.1 Background of the Study

Philippines is a typhoon-prone country in Southeast Asia. About 20 typhoons affect our country every year.

Typhoon Haiyan that made landfall at Eastern Samar in November 8, 2013 brought destructive storm surge. Haiyan developed into a Category 5 typhoon with wind speeds of 250 km/h that caused storm surge to elevate up to 5 meters. It brought devastating effects particularly in Tacloban City. More than 6,000 people are confirmed dead and 1.14 million houses damaged. Overall economic losses are estimated at 571.1 billion Pesos.

On top of that, surge height predictions from different agencies are not completely consistent. The JRC data indicate a storm surge South of the eye while Deltares sees this at the North of the typhoon. Deltares indicated maximum water height of 5 meters while JRC indicated about 2.3 meters. Project NOAH published its own data and indicated maximum water height of 5.16 meters. Thus, the country demands for a better and more accurate storm surge modeling.

1.2 Problem Statement

No model nor software has been tailored specifically for the Philippine setting to measure storm surge that occur every time a typhoon is in the county. Thus, the researchers are tailoring a model specifically for the Philippines and designing it to compute fast enough to be able to give real-time storm surge predictions around the country.

1.3 Objective

The research aims to tailor a model specifically for the Philippines and designing it to compute fast enough to be able to give real-time storm surge predictions around the country.

1.4 Significance of the Study

The research can have more realistic predictions at the least time. With this, disaster risk management will be more accurate.

1.5 Scope and Limitations

The will only use ADCIRC+SWAN for this research. There will be no changes in the underlying processes and source codes of these models. Changes will only be on the parameters and input files.

1.6 Definition of Terms

1.6.1 Geographical Concepts

Topography A detailed map of the surface features of land. It represents a particular area in detail, including everything natural and man-made hills, valleys, roads, or lakes. It's the geographical contours of the land.

Bathymetry The study of the "beds" or "floors" of water bodies, including the ocean, rivers, streams, and lakes; how deep they are.

Arcminute A unit of angular measure equal to 60 arcseconds, or 1/60 of a degree. The arc minute is denoted $'$ (not to be confused with the symbol for feet). In terms of length, 1 arcminute is equal to 1852 meters.

1.6.2 Geophysical Concepts

Storm Surge is an abnormal rise of water generated by a storm, over and above the predicted astronomical tides. This rise in water level can cause extreme flooding in coastal areas particularly when storm surge coincides with normal high tide, resulting in storm tides reaching up to 20 feet or more in some cases. A surge is generated by the storms spiraling winds and lower atmospheric pressure gathering a large volume of water along the surface and pulling it with the storm as it travels.

Barotropic is a condition where fluid density is only a function of pressure.

Baroclinic is another condition where fluid density is not only a function of pressure, but also temperature and salinity.

Coriolis Effect is apparent in the ordinary Newtonian laws of motion of bodies are to be used in a rotating frame of reference. An inertial force acts to the right of the direction of body motion for counterclockwise rotation of the reference frame or to the left for clockwise rotation. This must be included in the equations of motion (usually the Coriolis parameter in the momentum equations).

Coriolis Force is an apparent deflection of the path of an object that moves within a rotating coordinate system. The object does not actually deviate from its path, but it appears to do so because of the motion of the coordinate system.

Chapter 2

Review of Related Literature

2.1 Mechanics of Storm Surge

Tropical cyclones are circular storms characterized by high winds and heavy rainfall and form over warm, tropical oceans. The center of a cyclone is called the eye. It is surrounded by a ring of clouds called the eye wall, where the winds are strongest. Surrounding the eye wall are clouds that spiral outward, called spiraling rainbands.

A storm surge is primarily caused by the relationship between the winds and the oceans surface. The water level rises where the winds are strongest. In addition, water is pushed in the direction the winds are blowing. The rotation of the Earth causes winds to move toward the right in the Northern Hemisphere and toward the left in the Southern Hemisphere a phenomenon known as the Coriolis effect. If a cyclone develops in the Northern Hemisphere, the surge will be largest in the right-forward part of the storm. In the Southern Hemisphere, the surge will be largest in the left-forward part of the cyclone.

Another factor contributing to storm surge is atmospheric pressure. **Atmospheric pressure** is the force exerted by the weight of air in the Earths atmosphere. The pressure is higher at the edges of a cyclone than it is at the center. This pushes down the water in the outer parts of the storm, causing the water to bulge at the eye and eye wall where the winds have helped add to the rise in sea level.

Also called the **inverse barometer effect**, this is a result of the lower pressure of the storm causing the water to be displaced upward. The rule of thumb is for every 1 millibar (mb) drop in pressure there is a 1 centimeter (cm) rise in ocean surface.

Storm surge is caused primarily by the strong winds in a hurricane or tropical storm. The low pressure of the storm has minimal contribution. The wind circulation around the eye of a hurricane blows on the ocean surface and produces a vertical circulation in the ocean. In deep water, there is nothing to disturb this circulation and there is very little indication of storm surge.

Also, the central pressure of a hurricane is so low that the relative lack of atmospheric weight above the eye and eye wall causes a bulge in the ocean surface level. This effect is similar to using a straw. When you use a straw, you decrease the air pressure in the straw, and the high pressure pushing down on the rest of the drink pushes the drink up the straw. Going back to the scenario, it is the relative higher pressure on the ocean around the outside the hurricane that lifts the ocean surface in the center.

Once the hurricane reaches shallow waters near the coast, the vertical circulation in the ocean becomes disrupted by the ocean bottom. The water can no longer go down, so it has nowhere else to go but up and inland.

In general, storm surge occurs where winds are blowing onshore. The highest surge tends to occur near the **radius of maximum winds**, or where the strongest winds of the hurricane occur.

Total Water Level

In reality, storm surge only makes up a part of what causes water levels to rise along the coast during a hurricane. Here are the other factors:

Astronomical Tides

Water levels rise and fall along the coast every day due to the gravitational pull of the moon and sun, most commonly known as **tide**. When the tide is combined with the storm surge, it is called the **storm tide**. Usually it is safer to assume high tide when making decisions for risk assessment.

Waves

Breaking waves contribute to the water level rise through wave runup and wave setup. Wave runup occurs when a wave breaks and the water is propelled onto the beach. Wave setup occurs when waves continually break onshore and the water from the runup piles up along the coast because it cant get back out to sea. The water level therefore rises as a hurricane approaches, especially since the waves become larger and more water is pushed onshore.

Fresh Water Input

Heavy rainfall ahead of a hurricane can cause river levels to rise well inland from the coast. Once all this water flows downriver and reaches the coast, local water levels especially near deltas and bays will rise.

Water Level = Storm Surge + Tides + Waves + Freshwater Input

Factors that Influence Storm Surge

The maximum potential storm surge for a particular location depends on a number of different factors. Storm surge is a very complex phenomenon because it is sensitive to the slightest changes in storm intensity, forward speed, size which is attributed to radius of maximum winds or RMW, angle of approach to the coast, central pressure (minimal contribution in comparison to the wind), and the shape & characteristics of coastal features such as bays and estuaries.

Central Pressure

Lower pressure will produce a higher surge. However the central pressure is a minimal contributor compared to the other factors.

Storm Intensity

Stronger winds will produce a higher surge.

Size

A larger storm will produce higher surge. There are two reasons for this. First, the winds in a larger storm are pushing on a larger area of the ocean. Second, the strong winds in a larger storm will tend to affect an area longer than a smaller storm.

Storm Forward Speed

On the open coast, a faster storm will produce a higher surge. However, a higher surge is produced in bays, sounds, and other enclosed bodies of water with a slower storm.

Angle of Approach to Coast

The angle at which a storm approaches a coastline can affect how much surge is generated. A storm that moves onshore perpendicular to the coast is more likely to produce a higher storm surge than a storm that moves parallel to the coast or moves inland at an oblique angle.

Shape of the Coastline

Storm surge will be higher when a hurricane makes landfall on a concave coastline as opposed to a convex coastline.

Width and Slope of the Ocean Bottom

Higher storm surge occurs with wide, gently sloping continental shelves, while lower storm surge occurs with narrow, steeply sloping shelves.

Local Features

Storm surge is highly dependent on local features and barriers that will affect the flow of water. A good example is our country, which has the complexities of such features as barrier islands, inlets, sounds, bays, and rivers.

2.2 Advanced Multi-Dimensional Circulation Model (ADCIRC)

Developed as a joint project between the USACE Engineering Research and Development Center, University of Notre Dame, and University of North Carolina Chapel Hill [1], ADCIRC is a highly developed program for solving fluid motion equations on a rotating Earth. With equations formulated using traditional hydrostatic pressure and Boussinesq approximations and discretized in space using finite element and in time using finite difference, the result is a highly flexible non-structured grid. As a powerful tool for a variety of projects that use fluid motion modeling, it has an expanding user base and a number of supplementary software to better suit certain purposes. Being highly scalable with parallel computing, it was adopted later on by universities to provide supplementary surge forecasts. However, the power and accuracy of the model causes it to take more computing time relative to other surge models on the same scale.

ADCIRC is a system of computer programs for solving time dependent, free surface circulation and transport problems in two and three dimensions. These programs utilize the finite element method in space allowing the use of highly flexible, unstructured grids. Typical ADCIRC applications includes:

- Prediction of storm surge and flooding
- Modeling tides and wind driven circulation
- Larval transport studies
- Near shore marine operations
- Dredging feasibility and material disposal studies

Features of ADCIRC

ADCIRC is a highly developed computer program of Fortran 95 modules for solving the equations of motion for a moving fluid on a rotating earth. These equations have been formulated using the traditional hydrostatic pressure and Boussinesq approximations and have been discretized in space using the finite element (FE) method and in time using the finite difference (FD) method.

ADCIRC can be run either as a two-dimensional depth integrated (2DDI) model or as a three-dimensional (3D) model. In either case, elevation is obtained from the solution of the depth-integrated continuity equation in Generalized Wave-Continuity Equation (GWCE) form. Velocity is obtained from the solution of either the 2DDI or 3D momentum equations. All nonlinear terms have been retained in these equations.

ADCIRC can be run using either a Cartesian or a spherical coordinate system. The GWCE can be solved using either a consistent or a lumped mass matrix (via a compiler flag) and an implicit or explicit time stepping scheme (via variable time weighting coefficients). If a lumped, fully explicit formulation is specified, no matrix solver is necessary. In all other cases the GWCE is solved using the Jacobi preconditioned iterative solver from the ITPACKV 2D package.

The 2DDI momentum equations are lumped and therefore require no matrix solver. In 3D, vertical diffusion is treated implicitly and the vertical mass matrix is not lumped, thereby requiring the solution of a complex, tridiagonal matrix problem over the vertical at every horizontal node.

ADCIRC boundary conditions include:

- Specified elevation (harmonic tidal constituents or time series)
- Specified normal flow (harmonic tidal constituents or time series)
- Zero normal flow
- Slip or no slip conditions for velocity
- External barrier overflow out of the domain
- Internal barrier overflow between sections of the domain
- Surface stress (wind and/or wave radiation stress)
- Atmospheric pressure
- Outward radiation of waves (Sommerfield condition)

ADCIRC can be forced with:

- Elevation boundary conditions
- Normal flow boundary conditions
- Surface stress boundary conditions

- Tidal potential
- Earth load/self attraction tide

ADCIRC includes a least squares analysis routine that computes harmonic constituents for elevation and depth averaged velocity during the course of the run thereby avoiding the need to write out long time series for post processing. ADCIRC has been optimized by unrolling loops for enhanced performance on multiple computer architectures. ADCIRC includes MPI library calls to allow it to operate at high efficiency (typically better than 90 per cent) on parallel computer architectures.

2.3 Simulating Waves Nearshore (SWAN)

SWAN is a third-generation wave model, developed at Delft University of Technology, that computes random, short-crested wind-generated waves in coastal regions and inland waters[27].

Features of SWAN

SWAN can compute different physical phenomena involving like like the following:

- Wave propagation in time and space, shoaling, refraction due to current and depth, frequency shifting due to currents and non-stationary depth.
- Wave generation by wind.
- Three- and four-wave interactions.
- Whitecapping, bottom friction and depth-induced breaking.
- Dissipation due to aquatic vegetation, turbulent flow and viscous fluid mud.
- Wave-induced set-up.
- Propagation from laboratory up to global scales.
- Transmission through and reflection (specular and diffuse) against obstacles.
- Diffraction.

Computation

SWAN computations can be made on a regular, a curvilinear grid and a triangular mesh in a Cartesian or spherical coordinate system. Nested runs, using input from either SWAN, WAVEWATCH III or WAM can be made with SWAN.

SWAN runs can be done serial, i.e. one SWAN program on one processor, as well as parallel, i.e. one SWAN program on more than one processor. For the latter, two parallelization strategies are available:

- Distributed-memory paradigm using MPI and
- Shared-memory paradigm using OpenMP.

Output Quantities

SWAN provides the following output quantities (numerical files containing tables, maps and time series):

- One- and two-dimensional spectra,
- Significant wave height and wave periods,
- Average wave direction and directional spreading,
- One- and two-dimensional spectral source terms,
- Root-mean-square of the orbital near-bottom motion, dissipation,
- Wave-induced force (based on the radiation-stress gradients), set-up,
- Diffraction parameter, and many more.

Limitations

SWAN does not account for Bragg-scattering and wave tunneling.

2.4 Other Storm Surge Models

Storm Surge Modeling started out with empirically derived equations from existing storm surge data. One of the first of these equations was derived by Montgomery (1955) using surge data from the Gulf of Mexico. It took into account only the difference between the minimum pressure inside the storm and the ambient pressure to estimate the maximum surge height. From this as a starting point, researchers worked to make more accurate calculations by taking into account more factors found to have effects on the surge height such as wind speed and coastline shape. These historical methods of estimating surge height culminated in a computer-aided model developed by Harris (1963). His model took into account five factors: pressure effect, direct wind effect, Coriolis force, waves, and rainfall. It was observed that all five of these factors are directly related to pressure effects.

With further advances in computing power, more complex equations with more parameters came into use in modern programs used to model surge for emergency management and hazard mitigation.

Special Program to List Amplitudes of Surges from Hurricanes (SPLASH)

Released by Jelenianski et al. (1972) after improving upon empirical methods in the 1960s, SPLASH is an off-the-shelf program that takes into account pressure difference, maximum wind radius, coastal bathymetry and storm direction using three separate calculations. The first calculation uses pressure and wind radius in an equation based on earlier empirical methods to predict the maximum surge height. The next two calculations produce correction factors specific to a location to get the predicted surge for that location by multiplying the factors to the predicted maximum surge.

Tetra Tech (FEMA SURGE Model)

First introduced as TTSURGE by Tetra Tech (1976) in response to a recommendation by the National Academy of Science in 1975, it was extensively used and updated by FEMA so it eventually became known as FEMA SURGE. It was used by FEMA for Flood Rate Insurance Maps until the late 1980s and by coastal engineering firms in the early 1990s. It uses an explicit 2D space-averaged finite difference scheme and takes bathymetry, coastline configuration, boundary conditions, bottom friction and other flow coefficients as inputs for two models, a hurricane storm model and a hydrodynamic model. The computed wind stress from the hurricane storm model is then used as input for the hydrodynamic model.

Sea, Lake, and Overland Surges from Hurricanes (SLOSH)

Developed by the Techniques Development Laboratory of the National Weather Service (Jelesnianski et al., 1992) for use as a 2D numerical-dynamical model in real-time forecasting. It models the topography and obstacles on a curvilinear or polar grid (support for hyperbolic and elliptic grids were added later on) then calculates whether or not a grid cell will be inundated on a cell-by-cell basis. The model was kept relatively simple to keep the simulation runs short for its intended use in real-time forecasting.

Japan Meteorological Agency Model (JMA)

This model run eight times a day to provide predictions of storm tides along the Japanese coastline. In this attempt, the model predicts multiple scenarios of typhoons with different meteorological forcing fields to take account the uncertainty in typhoon track forecasts. The model computes only storm surges. Tides are computed separately using harmonic analysis and added to surge prediction after. Furthermore, it utilizes 2D shallow water form of momentum equations and the wave continuity equation. Nonlinear advection terms are omitted to increase computational efficiency and the equations are solved by numerical integration using explicit finite difference method.

2.5 Studies About Usage of ADCIRC+SWAN for Storm Surge Modeling

Real-Time Forecasting and Visualization of Hurricane Waves and Storm Surge Using SWAN+ADCIRC and FigureGen

In this paper, the authors presented an overview of the SWAN+ADCIRC modeling system for coastal waves and circulation. They also describe FigureGen, a graphics program adapted to visualize hurricane waves and storm surge as computed by these models. The system was applied recently to forecast Hurricane Isaac (2012) as it made landfall in southern Louisiana. Model results are shown to be an accurate warning of the impacts of waves and circulation along the northern Gulf coastline, especially when communicated to emergency managers as geo-referenced images.

Hurricane Gustav (2008) Waves and Storm Surge: Hindcast, Synoptic Analysis, and Validation in Southern Louisiana

Wave and surge computations from ADCIRC+SWAN are validated comprehensively at the locations ranging from the deep Gulf of Mexico and along the coast to the rivers and floodplains of southern Louisiana.

Coastal Tide Prediction Using the ADCIRC-2DDI Hydrodynamic Finite Element Model: Model Validation and Sensitivity Analyses In the Southern North Sea / English Channel

The selection of ADCIRC 2DDI finite element model is based on accuracy, mesh flexibility and computational efficiency. With respect to shallow water problems, the implementation of GWCE and momentum balance equations has well documented accuracy. The criteria of selection of ADCIRC is based on mass and momentum conservation, efficiency of the numerical solution algorithm and grid flexibility. Furthermore, due to extensive numerical testing and analysis of the model code, the efficiency and accuracy of ADCIRC is well understood.

Performance of the Unstructured-Mesh, ADCIRC+SWAN Model in Computing Hurricane Waves and Surge

Researchers from the University of Notre Dame examined the performance of the unstructured mesh and the coupled model applied to a high-resolution, 5M-vertex, finite element mesh of the Gulf of Mexico and Louisiana. The performance is compared to different platforms. SWAN and ADCIRC are coupled so that they run on the same computational core and on the same unstructured sub-mesh. The coupling paradigm maximizes efficiency in a parallel computing environment.

Chapter 3

Methodology

3.1 Machine Specification

This research employed the use of parallel computing to be able to more efficiently process the model. This research will use 10-node parallel computer network through a Beowulf cluster[6] built using 10 units of used AMD64 boxes Intel x86_64 architecture. Due to hardware limitations of the Beowulf cluster, the High Performance Computing grid of the Department of Science and Technology Advanced Science and Technology Institute was also used.

3.2 Input Files

Different input files are required by ADCIRC to run. The minimum requirements for it to run are the two input files: `fort.14` and `fort.15`. In line with the objectives, `fort.22` and `fort.26`, which are optional but needed in this research, will be included . Description of the input files are below.

3.2.1 Grid and Boundary Information File (`fort.14`)

This is required to run the ADCIRC model. This file will include the grid description, boundary type, finite element mesh of domain and the bathymetric depth of each node with respect to the geoid. This also includes elevation specified boundary forcing segments. More detailed description of this input file is given in [1]. To prepare the finite element mesh, the mesh generation software listed below was used.

Preparing Finite Element Mesh

1. The researchers use GEODAS Coastline extractor ETOPO1 to obtain the coastline and bathymetry data. The coastlines are accurate up to 1 arc-minute or 1,852 meters but only flat land topography is provided. Sperical coordinate system will be used in order to have the format in Lat-Long.

2. BATTRI for preparing finite element mesh. 401,584 nodes and 607,842 elements were generated.

3. `xmgredit`, OPNML and QGIS are used for visualizing mesh and computing boundary segments. There is only one set of boundary forcing segments.

4. Finally, combining the mesh and boundary segments will satisfy the requirements of `fort.14`. The file size is 52.8 MB.

5. `ADMAT` can verify `fort.14` if it has valid format and provide statistics about the grid file.

Mesh Refinement

Six mesh refinements are done to adhere with the numerical stability of ADCIRC.

1. Enforcing of Courant-Friedrichs-Lewy[11] condition over the bathymetry.

2. Imposing of a constant area constraint for elements shallower than 15 meters relevant to maximum depth in data.

3. Another area constraint related to wave propagation applied only to nodes deeper than 15 meters, again relevant to maximum depth in data.

4. Same with 3, however from 200 to 15 meters in depth only.

5. Last area constraint to help resolve some of the steep topography of some regions in the mesh.

6. Springs relaxation to generally improve quality of mesh, smoothing out the transitions from one element to another.

Along the creation, most of the cleaning and preprocessing of data is done manually. This includes deletion of some islands and also the use of QGIS for miscellaneous translations and general work on geographic data.

Assessment of the Grid file

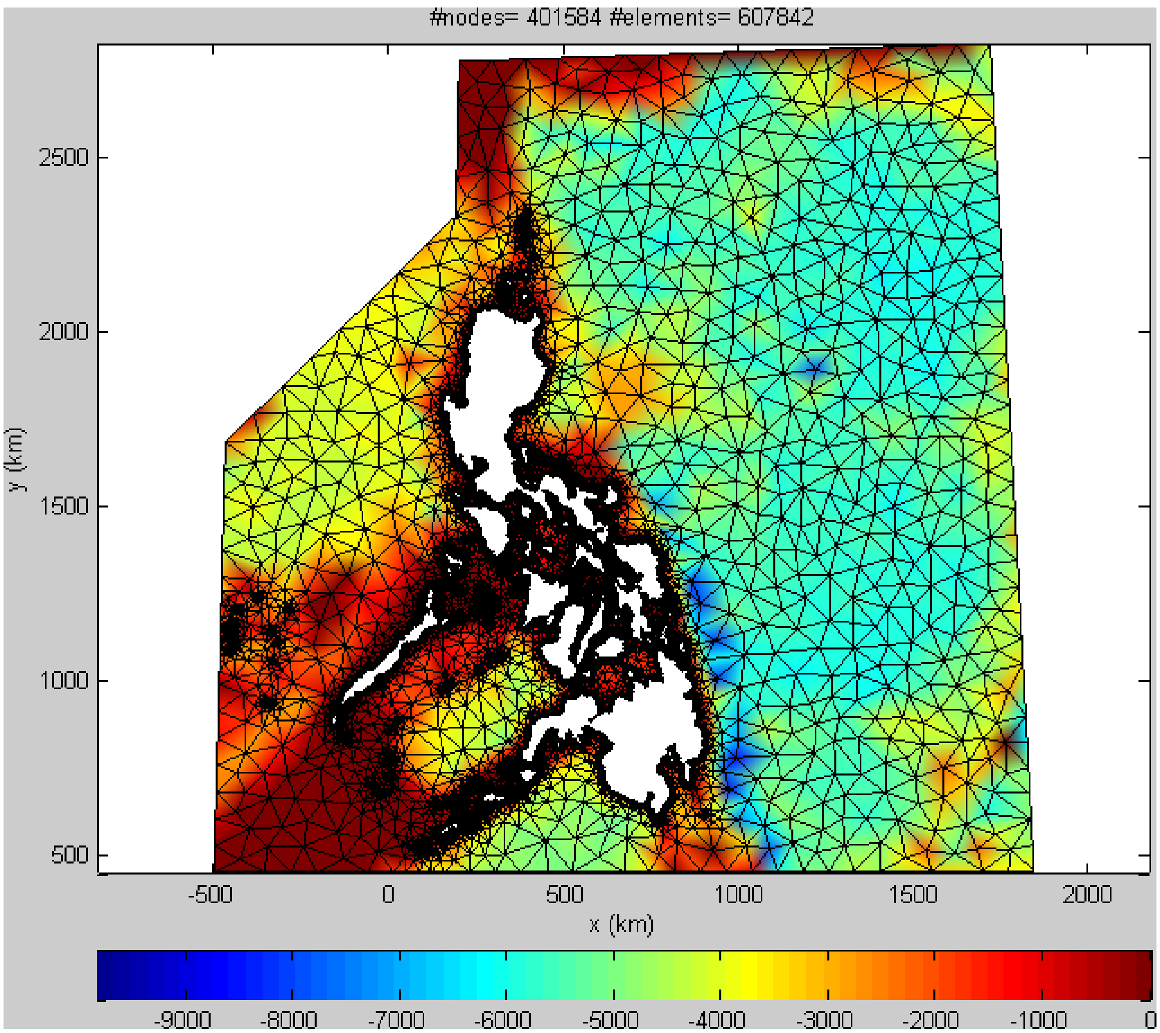

Figure 3.1: Philippine Area of Responsibility (PAR) with bathymetry.

The boundary is the Philippine Area of Responsibility (PAR) with Taiwan and Sabah removed. The Laguna de Bay is also removed temporarily. The bathymetry varies from 0 meters up to 9000 meters. It has 607,842 elements and 401,584 nodes. It is in the nodes where the computation is necessary.

Below is detailed view of finite element mesh of Samar and Leyte islands. As you can see, the mesh algorithm employs adaptive meshing - elements are coarse where computational results are not needed (like in the ocean) and elements are fine in the areas where computation is necessary (like in the shorelines). In the shorelines, the elements are fine so that the mesh adheres to the shoreline itself.

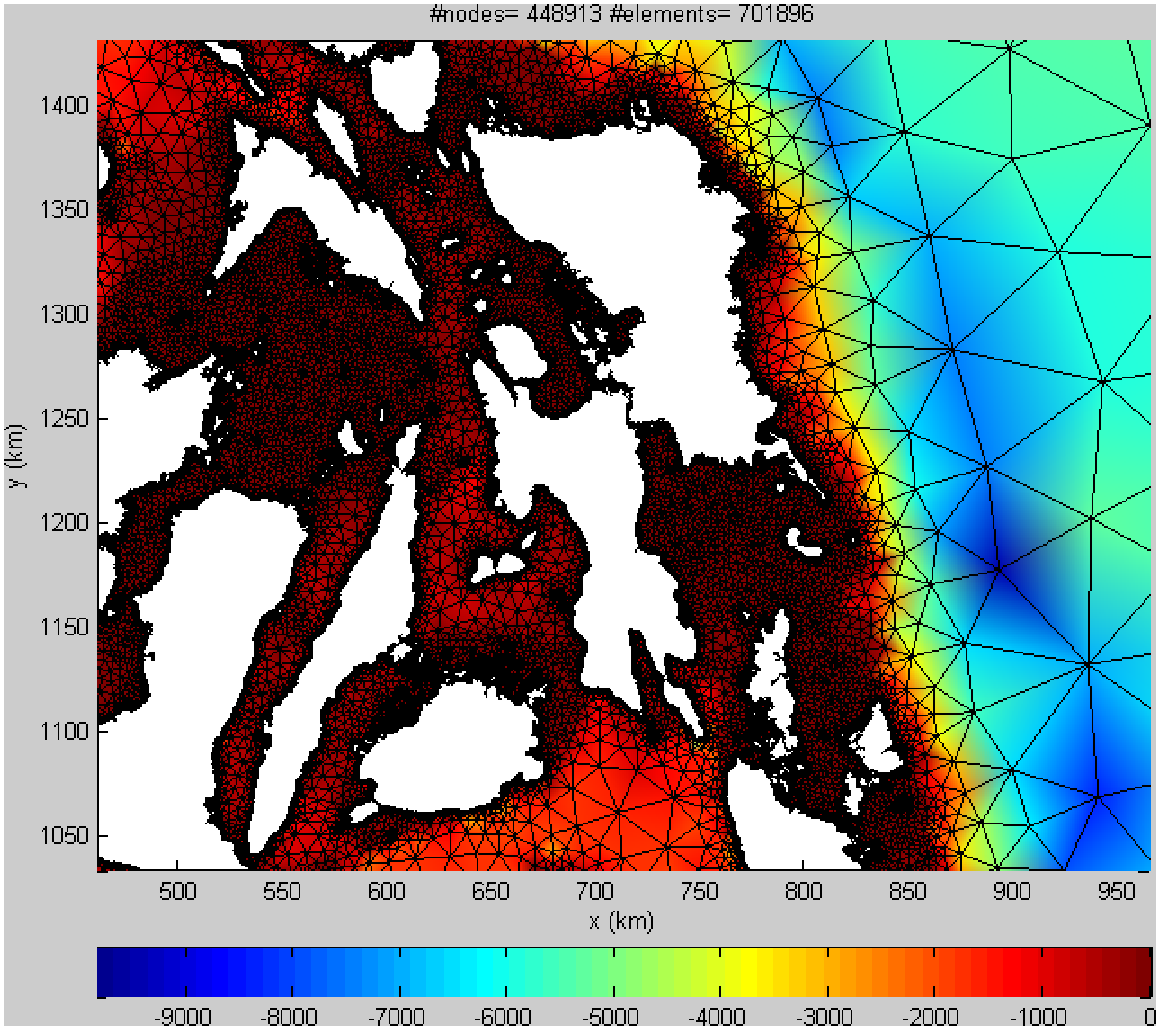

Figure 3.2: Samar and Leyte islands

Grid Quality

The quality of every element greatly contributes to the computing accuracy of ADCIRC+SWAN. It is the ratio of its shortest segment to its longest segment. An element is good for computation if its quality is greater than 0.6. If an element is an equilateral triangle, its quality is 1.0.

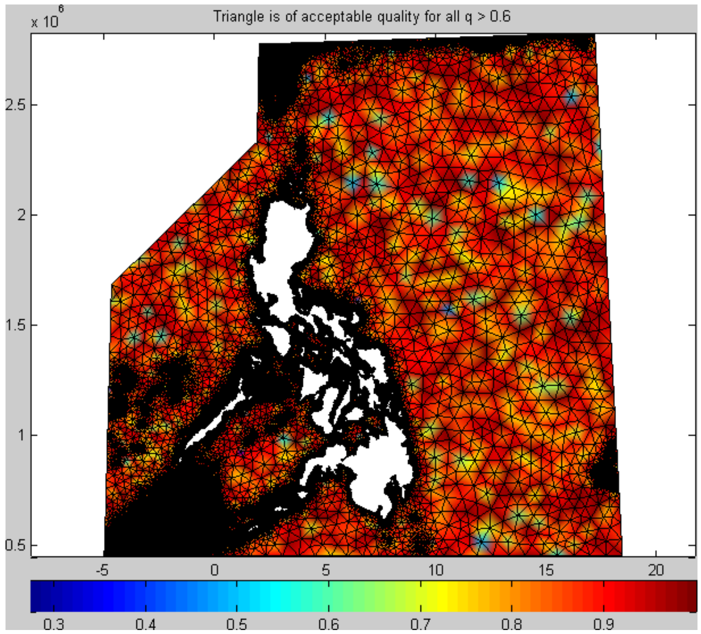

Figure 3.3: Distribution of element quality on the whole mesh.

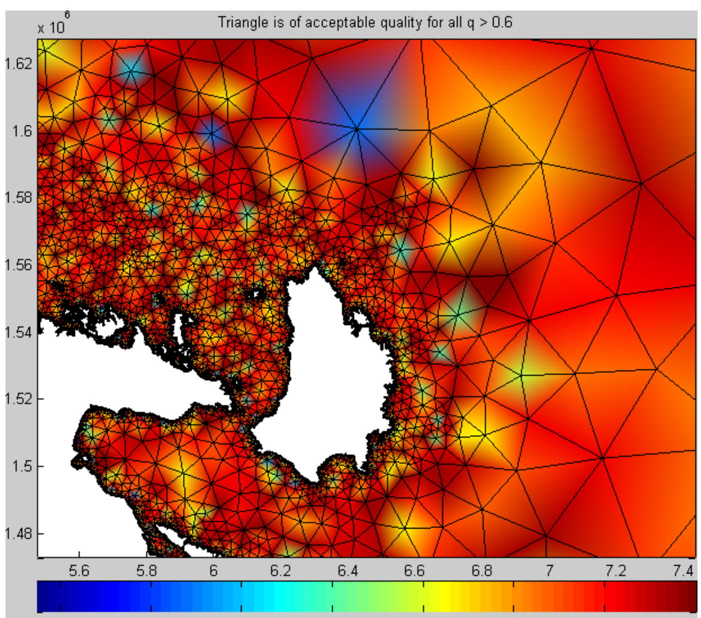

Figure 3.4: Problematic elements, near Catanduanes island

Statistics say that the grid file consists of 20,211 problematic elements. Most of the elements have good quality but poor elements can make the model unstable for computation.

ADMAT provided the following statistics about the grid file.

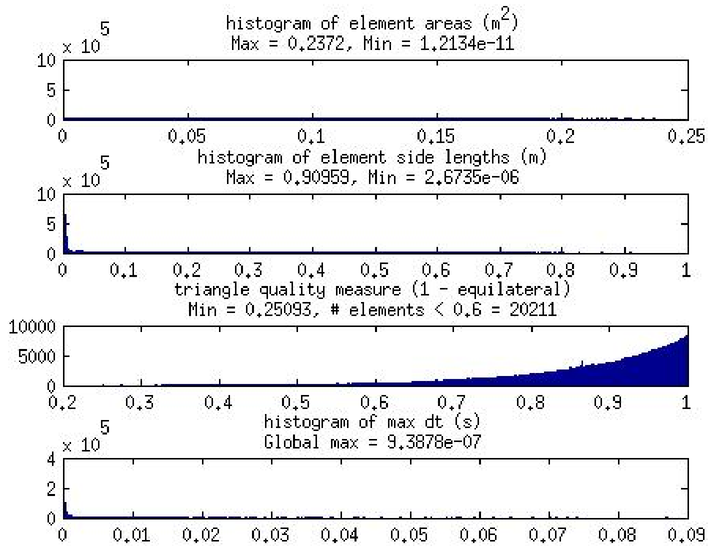

Figure 3.5: Statistics of `fort.14`

How the problem nodes can affect the accuracy of the model will be discussed in the succeeding chapters.

3.2.2 Model Parameter and Boundary Condition File (`fort.15`)

This file contains the majority of the parameters required to run both the 2DDI and 3D versions of ADCIRC and the information to drive the model with harmonic boundary conditions (either elevation or flux). This file also includes options on error logging, message logging, hot-starting the model, coordinate systems (Cartesian or spherical), density forcing, bottom friction, Coriolis parameter, tidal potential forcing, simulation time, reference time, time step, eddy viscosity, eddy diffusivity, gravitational constant and minimum velocity for wetting. It also provides options for including finite amplitude terms, advective terms, variable Coriolis parameter, meteorological forcing (related to `fort.22`) and elevation recording stations. The researchers also specify the model type to be used. This file is also required to run the ADCIRC model. Detailed structure of this file can be found in the documentation of `ADCIRC` [36].

The simulation used spherical coordinate system, constant Coriolis parameter and no open boundary forcing. Tidal potential constituents were also included. Simulation timeframe is 7 days from November 03, 2013 06:00 AM. The file size is 11.5 kB. Again, detailed specifications of this input file can be found in [1].

Coriolis Parameter

The user can define in `fort.15` if the model will have a constant or varying Coriolis parameter over time. A **Coriolis parameter** is defined as $f_c = 2\Omega\sin(\phi)$ where Ω is the frequency of the rotation of earth equal to 7.2921×10^{-5} rad/s and ϕ is the latitude. The user can define a spatially varying Coriolis parameter depending on the latitude of the current element included in the computation.

Tidal Potential Constituents

The user can also define in `fort.15` a set of tidal potential constituents[37] so that `ADCIRC` can simulate tides. For each tidal constituent, the user should define the descriptor, amplitude in meters, frequency in hertz, earth tide reduction factor, nodal factor and equilibrium argument in degrees. The descriptor, amplitude and frequency can be found on the website[37]. For the nodal factor and equilibrium argument, the user can use the Fortran program `tide_fac.f` that can be found in [23].

Building and Running `tide_fac.f`

Open the terminal and change the working directory to the one containing the code. Compile the code by issuing the command:

```
gfortran tide_fac.f -o tide.a
```

Then run using

```
./tide.a
```

And follow the instructions indicated. The user will put the starting date and the duration and the program will output the nodal factors and equilibrium argument in `tide_fac.out` file. To view its contents using the terminal, execute the command:
`cat tide_fac.out`

The following tidal constituents were used for both diurnal and semidiurnal periods:

TIPOTAG: descriptor	TPK: tidal potential amp in meters	AMIGT: frequency in Hz	FFT: nodal factor	FACET: eq. arg. in deg
K1	0.141565	0.00001160579658	0.91229	49.73
O1	0.100514	0.00001075849082	0.85654	105.85
P1	0.046843	0.00001154238062	1.00000	136.35
Q1	0.019256	0.00001033845625	0.85654	157.54
N2	0.046398	0.000021944146	1.03009	210.49
M2	0.242334	0.00002236428013	1.03009	158.80
S2	0.112841	0.00002314814815	1.00000	180.00
K2	0.030704	0.00002321139921	0.79692	278.61

Table 3.1: Tidal Constituents

It is recommended by ADCIRC that the earth tide reduction factor (ETRF) be 0.690[2].

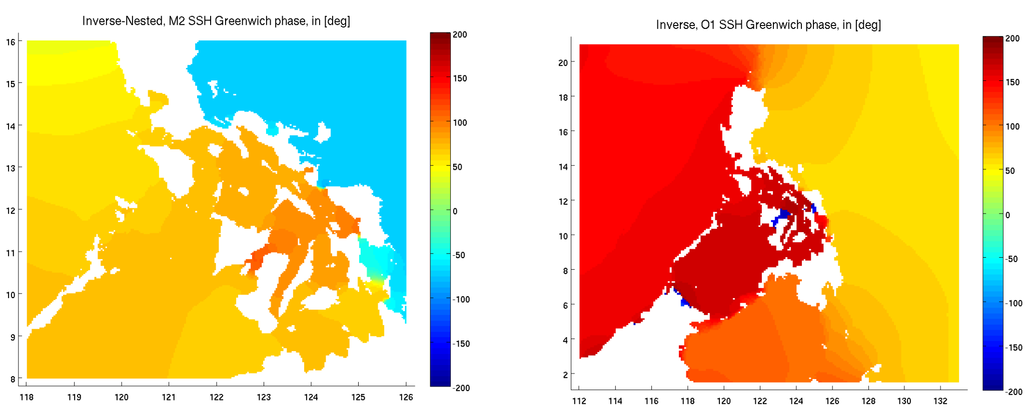

Figure 3.6: The tidal amplitude of M2 and O1 in the Philippines

3.2.3 Single File Meteorological Forcing Input (`fort.22`)

In `fort.15`, the option to include meteorological forcing and wind stress to the simulation was considered. These elements will be written in this file. ADCIRC requires that this file is in best track format that includes the coordinates of the eye, maximum sustained wind speed in knots, minimum sea level pressure in millibar, wind intensity in knots, radius of last closed isobar and background pressure. Storm data of Haiyan can be found in Global Disaster Alert and Coordination System (GDACS)[8]. After that, modification of columns 2 and 6 is necessary. Changed all entries of column 2 to 01, which determines the storm number that should agree on the storm number specified in fort.15. Also, change all entries from column 6 into 000, 006, 012, in increments of 6 forecast hours. The fize size is 4.8kB and detailed specifications can be found in [1].

3.2.4 SWAN Control File (`fort.26`)

This is the control file relevant to the coupling of SWAN to ADCIRC. This file includes information about how ADCIRC will send wind speeds, water levels and currents to SWAN. Detailed specification of this file can be found in [3].

3.2.5 Parameters Contributing to Computational Stability

Grid File

ADCIRC FAQ page recommends the following for stability:

- Compute the CFL parameter for each of the elements and ensure that it is less that 1 globally.
- Low quality elements (even just one or two) can give problems. Remember that if an element highly resembles an equilateral triangle, it is of good quality.

Model Parameter File

The paper Assessment of ADCIRC's Wetting and Drying Algorithm[32] suggests the following:

- The values of parameters V_{min}, τ and τ_0 defined in `fort.15` should vary depending on the linearity or nonlinearity of the bathymetry, bottom friction of the domain.
- Try running a test simulation with the non-linearity / wetting-drying features (`NOLIFA, NOLICA, NOLICAT, NRAMP`) turned off and then bring them back in one at a time, if you get a stable run.
- Try a very small timestep, $dt = 1sec$ is recommended.

In this research, $V_{min} = 0.05, \tau = 0.01$ and $\tau_0 = 0.01$.

3.3 Running the Model

The input files should be in the `work` directory of ADCIRC program. Running can be done in serial or in parallel. After running the program, output files will be generated on the same directory. It is recommended to timestamp the program execution for benchmarking purposes.

3.3.1 Serial Simulation

Preparing the Machine

In the terminal, proceed to the `work` directory of ADCIRC program and execute the commands

```
make clean
make clobber
```

to remove stray binary files.

```
make all
```

This command compiles the source codes and creates the executable files `adcprep`, `adcirc`, `padcirc` and `adcpost`. This will also display compilation errors, if there are any.

Building and Setting Up SWAN

From the `work` folder, do the following commands to compile the source codes relevant to SWAN.

```
cd ../swan         #to go to swan folder.
make clean && make clobber #to remove stray binary files.
make config  #to config swan
make punswan
make clean && make clobber
cd ../work #to go back to work folder.
make adcswan
```

Running the Model

When the input files are in the `work` folder, execute `./adcirc` on the same folder to begin the simulation if ADCIRC only, or `./adcswan` for ADCIRC+SWAN. The program will notify you of errors in the input files through the terminal if there are any. When it is finished, output files will be generated.

Coping With Hardware Limitations

Running serial ADCIRC+SWAN in the workstation with 8GB of RAM uses extensive memory and even swap memory. It produces segmentation fault error pointing to the address `0xffffffff` which might be the last memory address. In order to run the serial model, it demands the High Performance Cluster of the DOST-ASTI. After proper endorsement and gaining access, copy the whole project folder to your dedicated folder in the grid using secure copy command:

```
sudo scp -r <name of project folder><destination folder>
```

e.g. `sudo scp -r adcirc_par/ user@hpc:~`

After copying, repeat building and setting up the source codes. Note that building and setting up the source code may require administrative access, make sure to contact the system administrator.

3.3.2 Parallel Simulation

Setup for the Beowulf Cluster

The Beowulf Cluster was prepared as indicated in a paper by Curtis Michels, "Beowulf Clusters"[6]. It includes setting up the SSH access, OpenMPI and passwordless login for all nodes. Make sure to set static IP addresses.

Preparing the Server

In the terminal, proceed to the `work` directory of ADCIRC program and execute the commands

```
make clean
make clobber
```

to remove stray binary files.

Building and Setting Up ADCIRC

```
make all
```

This command compiles the source codes and creates the executable files `adcprep`, adcirc, padcirc and adcpost. This will also display compilation errors, if there are any.

Building and Setting Up SWAN

Use the following command to compile the source codes relevant to SWAN.

```
cd ../swan    #to go to swan folder.
make clean && make clobber    #to remove stray binary files.
make config #to config
make punswan
make clean && make clobber
cd ../work
make padcswan
```

Running the Model

Run ./adcprep
to prepare the Beowulf cluster for running parallel ADCIRC. This program distributes the grid files to different CPUs. To execute the model, execute the commands

```
mpiexec -hostfile <mpi_hostfile> -n <no_processors> ./padcirc -w <writer_cores>
#for ADCIRC only, or
mpiexec -hostfile <mpi_hostfile> -n <no_processors> ./padcirc -W <writer_cores>
#for ADCIRC+SWAN.
```

The hostfile tells ADCIRC which nodes are included in the model run. Just list all the IP addresses of the nodes in the cluster per line.

Note that the -W flag controls the dedicated writer cores, as described below. Using the dedicated writer cores is highly recommended. The idea is to set aside several cores for dedicated file output, thus freeing the remaining cores to continue with the computations. For example, you might run adcprep to decompose the global input files into 502 PE subdirectories. But the run itself would go on 512 cores, and the writer cores would be activated with the `-W` flag at run-time:

```
mpirun -np 512 ./padcswan -W 10
```

Significant speed-ups is noticeable when using the dedicated writer cores, and run-times can be cut in half or more on some machines. The user may set aside at least one writer core for each global output file. For SWAN+ADCIRC with all of the global output files being written, that means at least 10 nine writer cores are needed.

Postprocessing the Output Files

Run `./adcpost` to combine all output files from each node. Make sure to combine all relevant output files.

3.4 Output Files

Screen Output (`fort.6`)

During the program execution, limited run time information will be printed on the console where the program is run.

General Diagnostic Output (`fort.16`)

Output file which echo prints information from the Grid and Boundary Information File, the Model Parameter and Periodic Boundary Condition File, provides some processed information and prints out error messages from ADCIRC.

Iterative Solver Diagnostic Output (`fort.33`)

Diagnostic output from the ITPACKV 2D solver is written to this file if the solver encounters difficulty converging.

3D Density, Temperature and/or Salinity at Specified Recording Stations (`fort.41`)

3D Velocity at Specified Recording Stations (`fort.41`)

3D Turbulence at Specified Recording Stations (`fort.43`)

3D Density, Temperature and/or Salinity at All Nodes in the Model Grid(`fort.44`)

3D Velocity at All Nodes in the Model Grid (`fort.45`)

3D Turbulence at All Nodes in the Model Grid (`fort.46`)

Elevation Harmonic Constituents at All Nodes in the Model Grid (`fort.53`)
This output file contains amplitude and phase information obtained from harmonic analysis of the elevation solution at all nodes in the grid for the frequencies specified in the Model Parameter and Periodic Boundary Condition File.

Depth-averaged Velocity Harmonic Constituents at All Nodes in the Model Grid (`fort.54`)
This contains the amplitude and phase information obtained from harmonic analysis of the velocity solution at all nodes in the grid for the frequencies specified in the Model Parameter and Periodic Boundary Condition File.

Harmonic Constituent Diagnostic Output (`fort.55`)
This contains the comparison between the mean and variance of the elevation and velocity time series generated by the model and the mean and variance of a time series resynthesized from the computed harmonic constituents. This gives an indication of

how complete the harmonic analysis was.

Elevation Time Series at Specified Elevation Recording Stations (`fort.61`)

Depth-averaged Velocity Time Series at Specified Velocity Recording Stations (`fort.62`)

Elevation Time Series at All Nodes in the Model Grid (`fort.63`)
It contains elevation time series output at all nodes in the model grid as specified in the Model Parameter and Periodic Boundary Condition File.

Depth-averaged Velocity Time Series at All Nodes in the Model Grid (`fort.64`)
It outputs depth-averaged velocity time series output at all nodes in the model grid as specified in the Model Parameter and Periodic Boundary Condition File.

Maximum Elevation at All Nodes in the Model Grid (`maxele.63`)
It contains the maximum elevation output for each node in the grid for the entire model run.

Maximum Radiation Stress at All Nodes in the Model Grid (`maxrs.63`)
It contains the maximum radiation stress output for each node in the grid for the entire model run.

Maximum Water Velocity at All Nodes in the Model Grid (`maxvel.63`)
It contains the maximum water velocity output for each node in the grid for the entire model run.

Maximum Wind Velocity at All Nodes in the Model Grid (`maxwvel.63`)
It contains the maximum water velocity output for each node in the grid for the entire

model run.

Minimum Air Pressure at All Nodes in the Model Grid (`maxvel.63`)
It contains the minimum air pressure output for each node in the grid for the entire model run.

Atmospheric Pressure Time Series at Specified Meteorological Recording Stations (`fort.71`)

Wind Velocity Time Series at Specified Meteorological Recording Stations (`fort.72`)

Atmospheric Pressure Time Series at All Nodes in the Model Grid (`fort.73`)

Wind Stress or Velocity Time Series at All Nodes in the Model Grid (`fort.74`)
Wind velocity or stress time series output at all nodes in the model grid as specified in the Model Parameter and Periodic Boundary Condition File.

The user can choose if ADCIRC will produce these outputs on a user specified periodic timestep. Detailed format for the output files can be found at ADCIRC website[1].

SWAN Output Files

The user will also see output files from SWAN. These are:

- `rads.64` which contains the radiation stress gradients,
- `swan_HS.63` significant wave heights
- `swan_DIR.63` mean wave directions
- `swan_TMM10.63` mean wave periods

- `swan_TPS.63` peak wave periods

3.5 Graphical Output

The output files all encoded into plain text (ASCII). Converting the output files into graphical format is recommended to obtain meaningful results. There are two separate programs that can visualize ADCIRC output: FigureGen and ADMAT.

3.5.1 FigureGen

FigureGen is a Fortran program that creates images for ADCIRC files. It reads mesh files (`fort.14`, etc.), nodal attributes files (`fort.13`, etc.) and output files (`fort.63, fort.64`, maxele.63, etc.). It plots contours, contour lines, and vectors. Using FigureGen, you can go directly from the ADCIRC input and output files to a presentation-quality figure, for one or multiple time snaps.

Features

It also contains several new features:

- Compatibility with the NetCDF-formatted output files from SWAN+ADCIRC. These files are much smaller than the standard ASCII-formatted output files, and FigureGen can process them more quickly. This new feature is described below in Example 12.
- Capability to plot Lagrangian particles from the tracking code. This new feature was very useful in an oil spill modeling effort.
- Capability to plot background images behind the rest of the plot. This feature can be used to add any geo-referenced image.

System Requirements

FigureGen relies on other software as it makes images from ADCIRC-related files. You will need to install these software packages on your system before using FigureGen.

- Generic Mapping Tools (GMT) is the software package that plots the contours, contour lines, etc. and produces a PostScript file containing the figure. The GMT software package is actually a suite of executable programs, of which FigureGen uses only a subset. Note that FigureGen v.49 only works with GMT 4.5 or higher! `ps2rastercommand` was used to convert the PostScript files to raster images, and the ps2raster command was not implemented until GMT 4.5.
- GhostScript is the software package that converts PostScript files into different formats, including the raster format used by FigureGen. GhostScript is typically installed on Unix systems, so it may already be available on your system.
- ImageMagick is a software package for the manipulation and conversion of images. This software is optional! It is only required when FigureGen adds background images to the plots. In that case, the convert command needs to be available from ImageMagick. Otherwise, this software can be ignored.

The source code is `FigureGen49.F90`. This code will ask the user for and then read an input file containing information about which files to plot and how. FigureGen will plot the contours of both elevation-type (`fort.63, maxele.63, maxwvel.63`, etc.) and velocity-type (`fort.64, fort.74`, etc.) output files, mesh (`fort.14)` files, and nodal attributes (`fort.13)` files. The user can specify a color palette to use by creating and saving a palette in SMS. The user can convert units automatically and specify the lat/lon box to plot. And, perhaps most importantly, the user can run FigureGen in parallel, to create multiple images in a short amount of time.

Also, although many of the examples use different color palettes, it is best to start with the `Default2.pal` palette for most applications. This palette has been constructed painstakingly to contain the appropriate amount of purple. FigureGen will accept any color palette in SMS format, though, so you can feel free to experiment.

Building and Setting Up FigureGen

Serial compilation is fairly straight-forward; the following example uses the Portland Group compiler:

```
gfortran FigureGen49.F90 -o FigureGen49.exe
```

To compile for use in a parallel computing environment, you should add the -DCMPI pre-compiler flag. The following example uses the Intel compilers and MVAPICH2:

```
mpif90 FigureGen49.F90 -DCMPI -o FigureGen49.exe
```

Input File for FigureGen

Conventionally, the input file for FigureGen is named `input.inp`. In this file, the user will define parameters including plot label, time bar, latitude / longitude boxes, contours, scales, particles, vector flags, logo, image resolution and image output format.

Running FigureGen

The serial version of FigureGen can be executed from the command prompt:

```
./FigureGen49.exe
```

And the parallel version can be executed typically with a command like:

```
mpirun -np 10 ./FigureGen49.exe
```

depending on the system, of course. In both cases, FigureGen will ask the user for the name of the input file, and then it will proceed with the generation of the figure(s).

This new version 49 of FigureGen will also accept command-line arguments for the name of the input file and/or the version number. For example, the command:

```
./FigureGen49.exe -I input.inp -V 49
```

would instruct FigureGen that the input file is named `input.inp` and that it is formatted for version 49. FigureGen will read from this input file without prompting the user for the file name. This new feature can be useful in parallel computing environments.

3.5.2 ADMAT

ADMAT is a collection of Matlab scripts for the ADCIRC tidal model, Scripts are intended to aid in pre- and post-processing of ADCIRC related files. It reads output files of ADCIRC to make a graphical representation of its output. The output file of ADMAT is in AVI format. The user can produce output for a certain point in time or throughout the time frame. The ADMAT suite consist of the following MATLAB scripts:

1. `grid_quality.m` - this script will read in an ADCIRC grid (`fort.14` file) and make a number of 'diagnostic' plots.

2. `read_fort14.m` - this script will read in the grid from the `fort.14` file.

3. `read_fort15.m` - this script will read in the parameter data from the `fort.15` file.

4. `track_particles.m` - this script allows the user to specify initial locations for drifters and then, based upon a `fort.64` output file, computes the tracks taken

by the drifters.

5. `grid_lite.m` - this script is called by `track_particles.m` and is a 'faster' version of grid data.

6. `animate_elevation.m` - this script will read in a `fort.63` file and allow the user to make a static plot of elevation at a given time. Alternatively, the user can request an animation be made and saved as an `.avi` file.

7. `animate_tracks.m` - this script requires a previously saved .mat file containing drifter tracks computed from `track_particles.m`. Upon execution, this script will animate the motion of the drifters and save the results to an .avi file.

8. `plot_tidal_datums.m` - this script will make plots of tidal datums, such as MHHW, MLW, and so on. This script requires (i) that you perform a harmonic analysis (`fort.53`) files and (ii) convert this information into columns containing the various data.

9. `plot_rms_velocity.m` - this script essentially reads and plots the information in a `fort.54` file, which is the ADCIRC output for the velocity harmonic analysis. Specifically, the mfile plots the root mean square tidal speed throughout the domain.

10. `animate_velocity.m` - this script reads in velocity data (`fort.64` file) and allows the user to select between a plot at a single time or viewing / saving and `.avi` animation. Velocity vectors are plotted and the user can select between color contours of speed (most useful), bathymetry, and vorticity (not very useful to date).

11. `trigradient.m` - this script is called by `animate_velocity.m` in the instance where vorticity information is requested. It calculated derivatives for an irregular grid.

12. `elevation_timeseries.m` - this script basically reads and plots the information in a `fort.61` file. The user is allowed to select one or more of the stations at

which time series elevation data had been requested. The time series are then plotted.

Setting Up and Running ADMAT

After extracting the contents of ADMAT to a folder, add that folder to execution path of MATLAB. Execute any of the scripts given above to accomplish the required task and follow the instructions indicated.

Chapter 4

Results and Discussions

4.1 On ADCIRC + SWAN

Input files `fort.14`, `fort.15`, `fort.22` and `fort.26` are completed and run ADCIRC simulated on the specified timeframe. The table below lists the time and space considerations in running ADCIRC and ADMAT on serial using a computer with Intel(R) Core(TM) i3-2310M CPU @ 2.10 GHz running Ubuntu 11.04:

Case	Process Involved	Number of Timesteps	Time Requirement	Space Requirement (fort.63)
dt = 6hrs	ADCIRC	28	00:05:20	414.8Mb
	ADMAT		00:11:16	16.5Mb
dt = 3hrs	ADCIRC	56	00:09:58	813.6Mb
	ADMAT		00:20:35	33Mb
dt = 1hrs	ADCIRC	168	00:27:32	2.3Gb
	ADMAT		00:56:34	99.1Mb
dt = 0.5hrs	ADCIRC	336	00:52:32	5.0Gb
	ADMAT		01:48:34	180.1Mb

Table 4.1: Time and space requirement in running ADCIRC and ADMAT

Based on the simulation, when time = 4.5 days during typhoon Haiyan, the storm surge height in Tacloban area is approximately 5.5 meters. This result is in close approximation with actual measurements. However, there appear to be irregularities in specific areas such as surge persistence and absence of surge when these events are suspected to have happened. These irregularities may be attributed to instability of the computation caused by problem elements in the mesh.

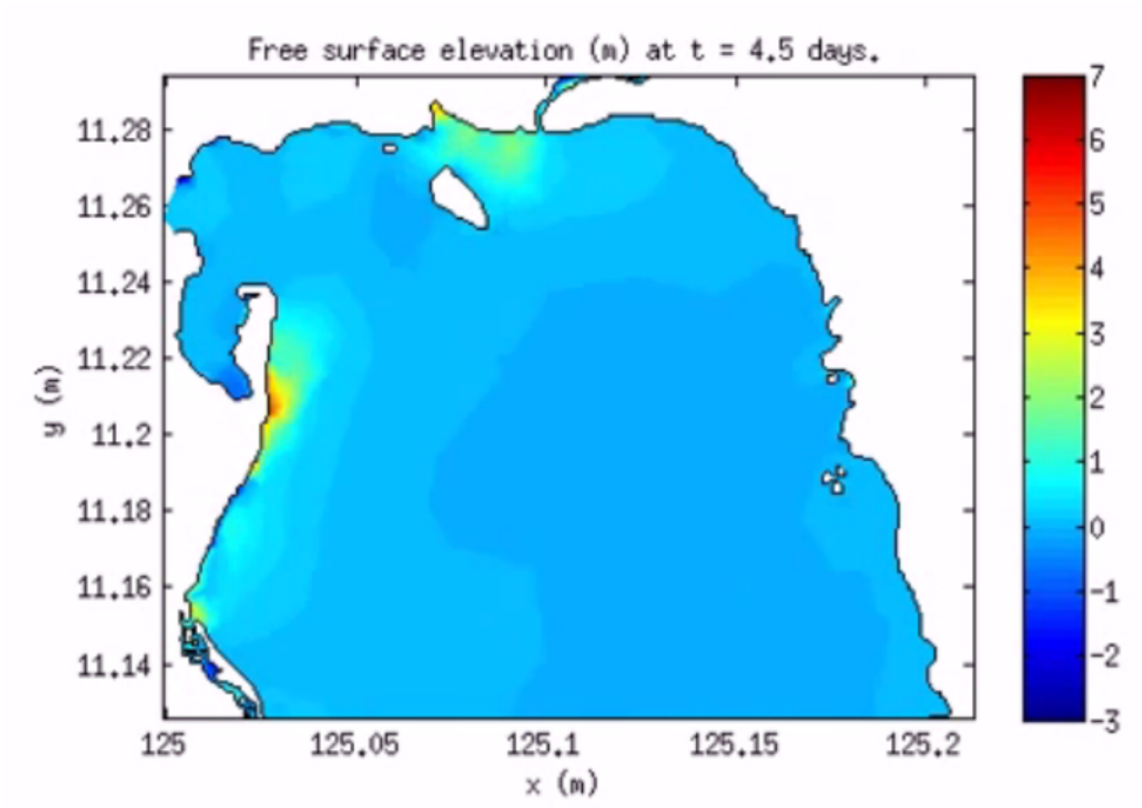

Figure 4.1: Predicted surface elevation at Tacloban Bay

Results on the predicted surge in Guiuan, Southeastern Samar near Philippine Trench was generated. When time = 4.5 days during the typhoon, predicted surge height is low, about 1.5 meters. This is also a good approximation with actual surge height. Remember that surge height is low when the bathymetry has steep slope from shallow to deep water.

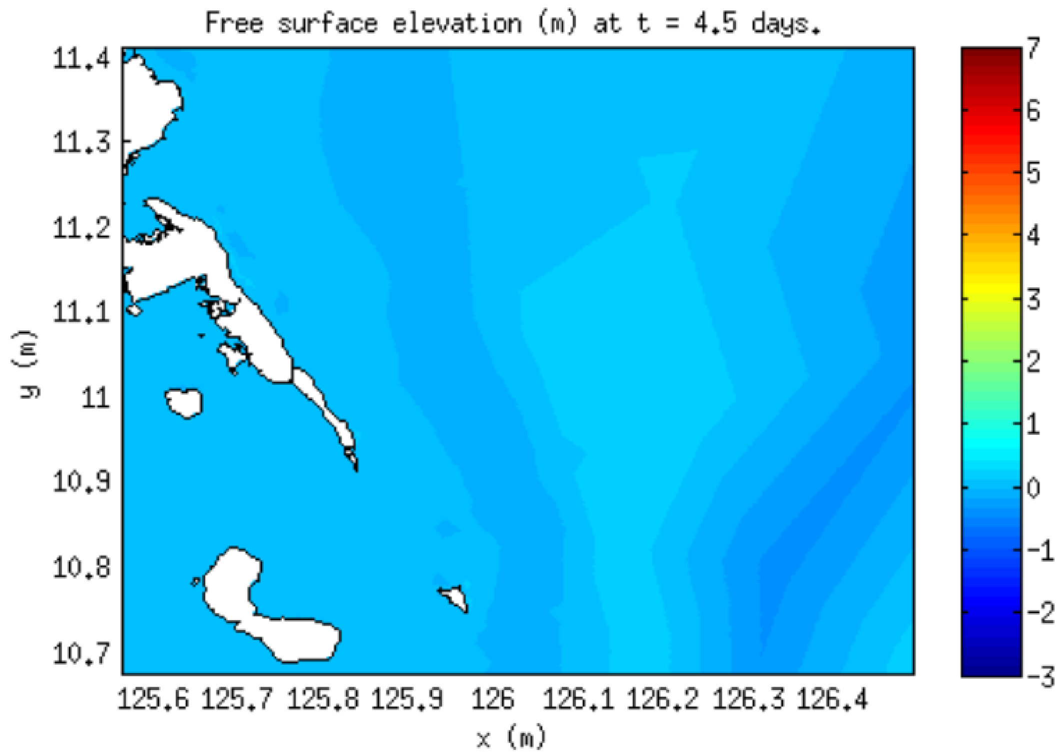

Figure 4.2: Predicted surface elevation at Guiuan, Southeastern Samar

4.2 On 3D ADCIRC + SWAN

The model is running smoothly, output files are generated but the need to visualize 3D output arises. Running time is approximately 5 times compared to 2D model run.

Chapter 5

Recommendations

To improve the simulation results and to make it finer, the researchers have the following recommendations:

- Run the 3D model that includes the nonlinearity in computation. This kind of run is much more resource-intensive but it may provide much neded accuracy and stability for the simulation.
- Explore other visualization tools such as FigureGen and StormSurgeViz. ADMAT only has tools for visualizing the water levels and water velocity of a 2D run. Other tools might be able to process the output of a 3D run and visualize other data such as pressure and wind speed.
- Have the model run with 1 second or sufficiently small timestep.
- Create `fort.14` without problem nodes and with 30 meters inland topography to determine the actual water inundation on land. This may be difficult on the full scale of the Philippine islands due to the sheer complexity of its coastline. It may be better to reduce the resolution of the mesh for the whole and generate finer meshes for specific areas as needed.
- Have the nodes in the Beowulf cluster share directories while running.
- In order to run the model in the Beowulf cluster, additional RAM is needed to avoid segmentation fault. Or, localize the mesh so that the memory consumption is low.

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